



SIMATS ENGINEERING

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Course Code: DSA0306

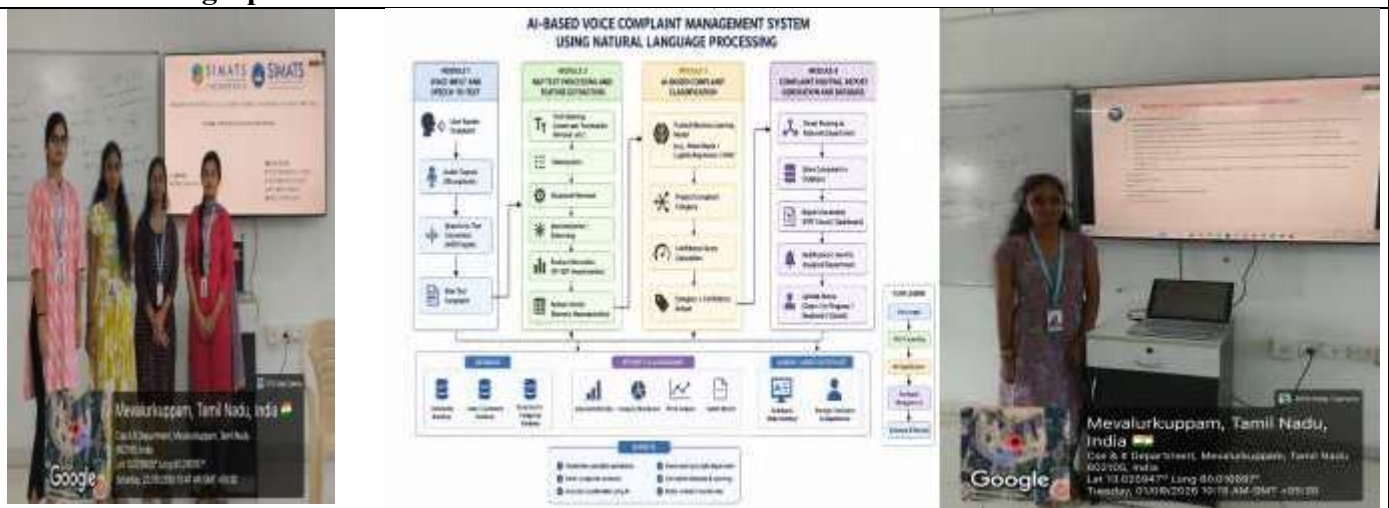
Slot: C

Course Name: Natural Language Processing

Course Faculty: Kumaragurubaran T

Project Title: Ai-Based Voice Complaint Management System Using Natural Language Processing

Module Photographs:



Project Description

The AI-Based Voice Complaint Management System Using NLP is an intelligent system designed to receive complaints through voice input and convert them into text using Speech-to-Text technology. The system applies Natural Language Processing (NLP) techniques to clean, process, and understand the complaint text.

AI-based classification is then used to automatically identify the complaint category, such as Network, Power, Maintenance, or General Services. This system reduces manual work and helps organizations manage complaints quickly and efficiently.

The system can be useful in colleges, hospitals, offices, customer service centers, and public service organizations. It can also be enhanced with priority detection, sentiment analysis, automatic department routing, and complaint report generation.

- **Module 1 – Voice Input and Speech-to-Text:** Captures the user's voice complaint and converts it into text.
- **Module 2 – NLP Text Processing and Feature Extraction:** Preprocesses the text and extracts important features using NLP techniques.
- **Module 3 – AI-Based Complaint Classification:** Classifies the complaint into the appropriate category using AI/ML techniques.

Student Signature

Guide Signature